## UME719: Refrigeration and Air Conditioning

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| 3 | 1 | 2* | 4.0 |

**Course Objective**: This course provides an introduction of different types of refrigeration systems and enables the students to analyze their performance using basic concepts of thermodynamics. This course also introduces the concept of psychometrics, air conditioning processes, air conditioning systems and refrigeration & air conditioning system components.

**Syllabus**

**Air and Vapour Compression Refrigeration:** Reversed Carnot cycle, air refrigeration cycle, aircraft refrigeration cycles, vapour compression refrigeration cycles, actual vapour compression cycle, advanced vapour compression refrigeration systems, compound compression and multi load systems, cryogenics refrigeration, cascade system and thermoelectric systems.

**Vapour Absorption Refrigeration:** Water vapour refrigeration systems, steam jet refrigeration; vapour absorption refrigeration systems, single effect and double effect vapour absorption systems.

**Refrigerants:** Desirable properties of common refrigerants, alternative refrigerants, refrigerator retrofitting procedure. Impact on environment by traditional refrigerants, refrigeration & associated equipment, ozone depletion and global warming.

**Refrigeration System Components:** Reciprocating Compressors and multistaging, expansion devices, condensers, evaporators, Compressors, expansion devices, condensers, evaporators.

**Air Conditioning:** Psychrometric properties of air, psychrometric processes, comfort charts, air conditioning load calculations, types of air conditioning systems. Demonstration of HVAC software related to psychometric processes & HVAC systems.

**Laboratory Work**

Experiments relating to measurement of performance parameters related to Refrigeration Bench, air conditioning test rig; Cold Storage Plant; Heat Pump Characteristics; Experimental Ice Plant; Cascade Refrigeration System; Rail Coach Air Conditioning Unit

**Course Learning Objectives (CLO)**

The students will be able to:

1. determine the COP for different types of air refrigeration systems
2. determine the COP for vapour compression systems and heat pump
3. perform thermodynamic analysis of absorption refrigeration systems and steam jet refrigeration system
4. perform the load calculations for the different type of air conditioning systems
5. identify and determine the heating and cooling loads for air conditioning systems

involving practical applications like; rooms/halls/restaurant/ theatre/auditorium etc.

**Text Books**

1. Domkundwar and Arora, Refrigeration and Air conditioning, Dhanpat Rai and Co, New Delhi (1973), 8thEdition, 2009.
2. Stoecker, W. F. and Jones J. W., Refrigeration and Air Conditioning, McGraw Hill, New York (1982).

**Reference Books**

1. Dossat, R. J., Principles of Refrigeration, Pearson Education, Singapore (2004).
2. Ameen, A., Refrigeration and Air Conditioning, Prentice Hall of India, New Delhi (2004).

**Evaluation Scheme**

| **S.No.** | **Evaluation Elements** | **Weightage (%)** |
|---|---|---|
| 1. | MST | 25 |
| 2. | EST | 45 |
| 3. | Sessionals (May include Assignments/Projects/Quizzes/Lab Evaluations) | 30 |